

§3 When is the ring just $\mathbb{Z}$?

only at $q = 2m + 1, 2m + 2$ (and $(6, 1)$)

§4 Which primes divide the index?

$p \mid q$ with $q' \mid h, h + 1, h + 2, q' \notin \{1, 2, 3, 4, 6\}$

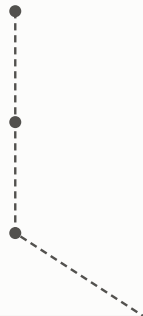
§5 What does the defect look like at p ?

layers W_i : lengths, exponent, Gorenstein, type

§6 Why do class numbers appear?

the moment matrix of the first layer

running example
 $A_3(35)$



6.1 sawtooth

Sinnott's index, doubling, h^-

6.2 shifted sawtooth

factor at 2: is $-1 \in \langle 2 \rangle$?

§7 What lies beyond the first layer?

universal ladder below p^a , descent at p^a ; certified exponents

§8 conclusions and limits · §9 how the numbers were computed